# Kinetics and Mass-Transfer Phenomena in Anaerobic Granular Sludge

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**Abstract:** The kinetic properties of acetate-degrading methanogenic granular sludge of different mean diameters were assessed at different up-flow velocities ($V_{up}$). Using this approach, the influence of internal and external mass transfer could be estimated. First, the apparent Monod constant ($K_S$) for each data set was calculated by means of a curve-fitting procedure. The experimental results revealed that variations in the $V_{up}$ did not affect the apparent $K_S$-value, indicating that external mass-transport resistance normally can be neglected. With regard to the granule size, a clear increase in $K_S$ was found at increasing granule diameters. The experimental data were further used to validate a dynamic mathematical biofilm model. The biofilm model was able to describe reaction-diffusion kinetics in anaerobic granules, using a single value for the effective diffusion coefficient in the granules. This suggests that biogas formation did not influence the diffusion-rates in the granular biomass. © 2001 John Wiley & Sons, Inc. *Biotechnol Bioeng* **73:** 125–134, 2001.

**Keywords:** mass transport; biofilm; kinetics; anaerobic; EGSB; granules

## INTRODUCTION

Today, anaerobic wastewater treatment is a well-accepted technology. This is mainly due to the development of high-rate reactors. High-rate reactors have the capacity to uncouple solid- and liquid-retention times resulting in high concentrations of biomass. The high-biomass concentrations are accomplished either by means of bacterial growth on an inert carrier as in fluidized-bed (FB) and fixed-film (FF) reactors, or by self-immobilization of biomass in the form of granules in upflow anaerobic and expanded granular sludge-bed reactors (UASB and EGSB, respectively).

Thick biofilms may give rise to mass-transfer limitations, resulting in an overall limitation of reactor capacity. In these conditions, the influx of substrate and/or outflux of products may become the rate-determining step. Mass-transfer limitations are particularly important at low-substrate concentrations, which are normally maintained in well-mixed reactors with low-effluent concentrations, such as EGSB-reactors. Mass-transport phenomena have hardly been studied in anaerobic systems. Contradicting results have been described regarding the impact of external and internal mass transport. While some reports indicate that both internal and external diffusion-limitations influence the rate of substrate utilization (Dolfing, 1985; Wu et al., 1995); others indicate that mass-transport limitations are not observed in anaerobic biofilms (de Beer et al., 1992; Schmidt and Ahring, 1991) even at a film thickness of 2.6 mm (Kennedy and Droste, 1986). In addition, it has been concluded that pH profiles inside anaerobic granules may affect the conversion rate more than mass-transport limitations (de Beer et al., 1992). The impact of diffusion and mass-transfer resistance on substrate utilization has been studied extensively in aerobic systems where clearly oxygen can be externally and internally mass-transport limited due to its poor water solubility (de Gooijer et al., 1991; Wijffels et al., 1991). However, in the case of anaerobic systems controversies still exist with respect to: (1) Whether diffusion can be rate-limiting in anaerobic biofilms, (2) whether increased liquid-upflow velocities ($V_{up}$) enhance mass transport in granular sludge, and (3) the relative importance of external and internal mass-transport limitations in anaerobic granules.

Several biofilm models have been developed that comprise the effects of mass transfer in anaerobic films. They include steady-state simulations of substrate profiles for multitrophic films with homogeneous (Buffiere et al., 1995) and heterogeneous biomass distribution (Arcand et al., 1995; Gupta et al., 1997; Tartakovsky and Guiot, 1997). Other models also include pH effects on kinetics (Flora et al., 1993, 1995). However, detailed experimental data to validate the models is limited.

The objective of this study was to investigate under which conditions anaerobic granular sludge could be mass-transport limited and if increased $V_{up}$ could enhance mass transport. In addition, the relative importance of external and internal mass-transport limitations in anaerobic granules was assessed.

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## MATERIALS AND METHODS

### Inoculum

Anaerobic granular sludge from a full-scale expanded granular-sludge bed (EGSB) reactor (Heineken, Zoeterwoude, The Netherlands) treating brewery wastewater was used. Both granular and dispersed granular sludge were used. Granular sludge of different particle sizes was obtained by sieving the sludge through metallic sieves under water. The dispersed sludge was obtained by disrupting the granular structure with a blender (Braun 4164).

### Assessment of Kinetic Parameters of the Inoculum

The maintenance coefficient (*Ms*) is defined as the quotient of the decay rate over the yield coefficient ($Ms = k_d/Y_{XS}$) as described by Leenen et al. (1996). By assessing the decay rate (and providing the yield is known), the maintenance coefficient can be estimated. The decay rate of dispersed granular biomass was assessed by determining the loss of specific methanogenic activity in absence of substrate. Twenty-four serum bottles of 120 mL were prepared by adding to each bottle, 2 g VSS/L and 50 mL of medium. This medium contained the following components in g/L of demineralized water: $NaHCO_3$ 4, $NH_4Cl$ 0.28, $K_2HPO_4$ 0.25, $MgSO_4 \cdot 7H_2O$ 0.1, $CaCl_2 \cdot H_2O$ 0.01, yeast extract 0.1, and trace elements 1 mL/L with the composition described below. The bottles were flushed with $N_2$ and $CO_2$ [70:30 (vol/vol)] and incubated at 30°C without substrate in a shaker. At 0, 3, 7, 14, 21, 35, 70, and 180 d, three bottles were taken, and the headspace was flushed with the same gas mixture, and 1 g acetate-COD/L was added to each bottle. The bottles were incubated again, and the methane in the headspace was measured over time. The maximum specific activity ($A^{max}$) was obtained from the slope of the steepest part of the cumulative-methane vs. time curve. The decay coefficient ($k_d$) was obtained from the slope of the ln ($A^{max}(t)/(A^{max}(0))$ vs. time curve. The correlation coefficient (i.e., $R^2$) of the straight line obtained was 0.96. The maximum growth rate ($\mu^{max}$) was assessed in a continuously stirred-batch reactor operated at 30°C, dispersed granular sludge at a concentration of 0.5 g VSS/L, and 2.5 L of mineral medium of the composition described below were placed in the system. The reactor was flushed with 100% $N_2$ for 15 min, then 2 g acetate-COD/L was added. Over 11 d, substrate was added daily to keep the concentration in the reactor between 1.5 to 2 g acetate-COD/L, and the methane production was measured on-line as described below. For zero-order kinetics ($S >> K_S$), the Monod-based equation for product formation, in this case methane, can be integrated (neglecting maintenance and/or decay), and the following equation applies:

$$C_{M,R}(t) = C_{M,R}(0) - f_{S,M} \cdot (1 - Y_{XS}) \cdot \frac{C_{X,R}(0)}{Y_{XS}} \cdot (1 - e^{\mu^{max} \cdot t}) \quad (1)$$

Where, $f_{S,M}$ refers to the COD-based stoichiometry for the conversion of substrate into product (methane). By using the biomass yield on acetate reported by Haandel and Lettinga (1994), and the on-line measurement of the methane in the reactor ($C_{M,R}$) as a function of time, the initial biomass concentration ($C_{X,R}(0)$), and $\mu^{max}$ were estimated with an optimization procedure included in a spreadsheet program. The optimization was based on minimizing the absolute error between measured and calculated values of methane.

### Mineral Medium

The medium used contained the following components in g/L of demineralized water: $NH_4Cl$ 0.28, $MgSO_4 \cdot 7H_2O$ 0.1, $KH_2PO_4$ 6.79, $Na_2HPO_4 \cdot 2H_2O$ 8.89, yeast extract 0.1, trace elements 1 mL/L. The trace elements solution contained the following components in mg/L of demineralized water: $FeCl_2 \cdot 4H_2O$ 2000, $H_3BO_3$ 50, $ZnCl_2$ 50, $CuCl_2 \cdot 2H_2O$ 38, $MnCl_2 \cdot 4H_2O$ 500, $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$ 50, $AlCl_3 \cdot 6H_2O$ 90, $CoCl_2 \cdot 6H_2O$ 2000, $NiCl_2 \cdot 6H_2O$ 142, $Na_2SeO \cdot 5H_2O$ 164, EDTA 1000, Resazurine 200, HCl 36% 1 mL/L. All chemicals were of analytical grade and purchased from Merck (Darmstadt, Germany); resazurine was from Fluka (Buchs, Switzerland) and yeast extract from Oxoid Unipath Ltd. (Hampshire, England).

### Preliminary Studies on the Kinetics of Dispersed Sludge Versus Granular Sludge

The kinetics of acetate degradation by crushed and granular sludge were determined by on-line measurement of the methane production rate in continuously stirred-batch reactors of 2.5 L. The reactors were filled with medium and inoculated with 2 g VSS/L of dispersed or granular biomass. Acetate was used as substrate at an initial concentration of 2 g COD/L. With the phosphate buffer added, the pH was kept at 6.8–7.2 and the reactors were operated in a temperature-controlled (30 ± 1°C) room.

### Kinetics of Granular Sludge

The characteristics of acetate degradation by anaerobic granular sludge of different diameters and at different $V_{up}$ values were determined by on-line measurement of the methane production rate in batch-recirculated systems. A schematic representation of the experimental set-up is shown in Figure 1. The system consisted of a glass reactor of 0.16 L and substrate was recirculated over the reactor continuously from a batch plastic vessel of 5 L. The system was filled with mineral medium and the reactors were seeded with 42.5 g of $VSS/L_{Reactor}$, of granular sludge of the specific mean diameter to be tested. The granules were left to expand according to the $V_{up}$ applied. To avoid biomass wash-out from the reactor, metallic sieves were placed at the bottom and top of the expanded sludge bed. The system was then flushed with 100% $N_2$ for about 15 min. Thereafter, subsequent feedings of 2 g acetate-COD/L were applied.



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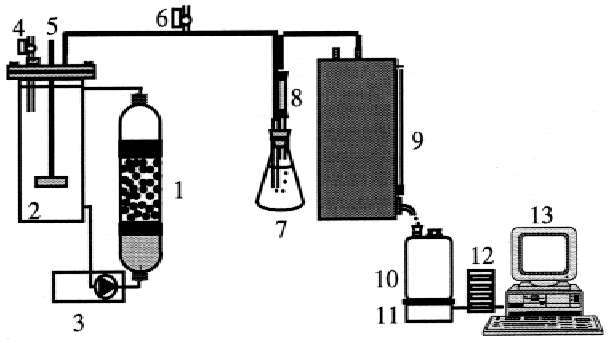



**Figure 1.** Experimental set-up for the assessment of the kinetics of granular sludge at different upflow velocities. (1) Glass reactor, (2) recirculation vessel or buffer tank, (3) recirculation pump, (4) liquid sampling port, (5) stirrer, (6) gas-sampling port, (7) NaOH washing bottle, (8) lime soda pellets, (9) Mariotte flask, (10) plastic container, (11) balance, (12) data logger, (13) personal computer. The set-up was the same for the kinetics of crushed versus granular sludge except that the glass reactor and the pump were not present; the reactor was the buffer tank.

After the second feed, substrate was added as acetic acid, which was neutralized by the bicarbonate produced in the previous feed. After each feed with acetic acid, the reactors were flushed with 100% $N_2$ for 15 min to remove the $CO_2$ formed by the chemical reaction. Every new feeding was added after the former feeding had been depleted. A complete run consisted of four feedings. The phosphate buffer added kept the pH at 6.8–7.2. The reactors were operated at 30 ± 1°C. All assays were conducted in duplicate.

### On-Line Methane Production Monitoring

The biogas produced was passed through an Erlenmeyer flask filled with 20% NaOH solution and then through a tube filled with soda lime pellets with thymol blue indicator. Finally, the gas passed through a Mariotte flask system filled with water for quantification of the methane produced. The methane production was monitored continuously by measuring the weight of the displaced water with a pressure sensor (model QB745, DS-Europe) connected to a programmable data logger system (Model CR10, Campbell). The data were recorded every 10 s and averaged over a 15-min interval. To communicate with the data logger, a personal computer programmed to function as terminal emulator was used. In all experiments the amount of methane produced corresponded (on COD-basis) to 95–108% of the amount of substrate added, thus justifying the experimental approach utilized.

### Analytical Methods

The concentration of volatile fatty acids in the medium was determined with a gas chromatograph (Hewlett Packard 5890A). A glass column (2 m × 6 mm) packed with Supelcoport (100 to 120 mesh), coated with 10% Fluorad FC 431, was used. The temperatures of the column, injection port and flame ionization detector were 130, 200, and 280°C, respectively. Nitrogen gas saturated with 100% formic acid was used as a carrier gas at a flow rate of 40 mL/min. Methane was analyzed with a gas chromatograph (Hewlett Packard 438/S), equipped with a steel column (2 m × 2 mm), packed with Poropak Q (80 to 100 mesh). The temperatures of the column, injection port, and flame ionization detector were 60, 200, and 220°C, respectively. All gas samples were taken with a gas-tight syringe (Dynatech, Baton Rouge, LA). The concentration of volatile suspended solids (VSS) was determined by standard methods [American Public Health Association (APHA), 1985]. The mean diameter of the granular sludge was determined before and after every experimental run with an image analyzer. Sludge samples were placed in a 6-mm petri dish; pictures of the samples were made with a camera (Sony CCD) equipped with a 55-mm micro Nikkon lens. The pictures were digitalized and analyzed with a Magiscan 2 Image Analysis System (Applied Imaging, Tyne & Wear, UK) using the image-analysis software package GENIAS v. 4.6 (Applied Imaging Tyne & Wear, UK). By sieving the dispersed sludge through metallic sieves, it was determined that its size was smaller than 200 μm.

## MATHEMATICAL MODEL DESCRIPTION

Two independent mathematical models were used to analyze the experimental results: A Monod-based model for calculation of the apparent $K_S$-values of the granular sludge under different operational conditions, and a biofilm model was developed to validate reaction-diffusion kinetics.

### Monod Model

Based on Monod kinetics, the rate of substrate consumption ($R_{S,R}$), biomass growth ($R_{X,R}$), and product formation ($R_{M,R}$) per unit of volume reactor in a batch reactor can be described by the following equations (Beeftink et al., 1990):

$$R_{S,R} = -\left(\frac{\mu^{max}}{Y_{XS}} + Ms\right) \cdot \left(\frac{C_{S,R}}{C_{S,R} + K_S}\right) \cdot C_{X,R} \qquad (2)$$

$$R_{X,R} = \left\{\mu^{max} \cdot \frac{C_{S,R}}{C_{S,R} + K_S} - Ms \cdot Y_{XS} \cdot \left(1 - \frac{C_{S,R}}{C_{S,R} + K_S}\right)\right\} \cdot C_{X,R} \qquad (3)$$

$$R_{M,R} = -R_{S,R} - \mu^{max} \frac{C_{S,R}}{C_{S,R} + K_S} \cdot C_{X,R} \qquad (4)$$

Where $\mu^{max}$ is the maximum growth rate, $K_S$ is the Monod half-saturation constant, $Ms$ is the maintenance coefficient, $Y_{XS}$ is the biomass yield, and $C_{S,R}$ and $C_{X,R}$ stand for the substrate, and biomass concentration in the reactor, respectively.

By combining these kinetic equations with mass balances for acetate conversion, the differential equations can be derived. These equations were integrated by using a fourth-order Runge-Kutta algorithm with adaptive step-size con-



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trol. The $K_S$-value, and the initial substrate [$C_{S,R}$ (0)] and biomass concentrations [$C_{X,R}$ (0)] were estimated for each substrate feeding by minimizing the absolute error between measured and calculated values of the methane-production rate. The estimated final biomass concentration from the previous feed was used as input value for the computations of the next feed. The parameter $K_S$ and initial values $C_{S,R}$ (0) and $C_{X,R}$ (0) are strongly independent, therefore they can readily be estimated in a single optimization procedure. The $K_S$-value is strictly related to the transition period in the methane production rate curve, the $C_{S,R}$ (0)-values are only dependent on the final cumulative amount of methane measured, and the $C_{X,R}$ (0)-is strictly dependent on the initial methane-production rate. For the minimization procedure, the Downhill Simplex method by Nelder and Mead (Press et al., 1989) was used. The estimated values for $C_{S,R}$ (0) and $C_{X,R}$ (0) were later used as input for the reaction-diffusion model.

## Biofilm Model

### General Assumptions

The biofilm model here described is subject to the following basic assumptions: (1) The granules are spherical in shape and uniform in size (for each test run, granules were size selected), (2) the number of granules and the size of the granules are constant in time, (3) for each experimental run, the $V_{up}$ is considered constant, (4) only radial diffusion transport is considered and is described by Fick's law, (5) the diffusion coefficient is constant, (6) there are no active biomass gradients in the granular biomass at time zero, and (7) the biological reactions are described according to the model suggested by Beeftink et al. (1990).

### Model Description

Discretization in time of the partial-differential equation describing reaction-diffusion kinetics in a spherical particle (i.e., granule) results in the following equation:

$$\frac{d^2C_{S,P}(r)}{dr^2} = -\frac{2}{r}\cdot\frac{dC_{S,P}(r)}{dr} + \frac{R_{S,P}}{\alpha\cdot D_S} \tag{5}$$

Where $C_{S,P}$ is the substrate concentration in the particle, $R_{S,P}$ is the volumetric substrate conversion rate in the particle, $\alpha$ is the relative effective diffusivity, and $D_S$ is the diffusion coefficient of substrate in water. The two boundary conditions to solve Eq. (5) are:

$$\left.\frac{dC_{S,P}}{dr}\right|_{r=0} = 0$$

$$C_{S,R} = C_{S,P}\big|_{r=Rp} + \frac{\alpha\cdot D_S}{k_1}\cdot\left.\frac{dC_{S,P}}{dr}\right|_{r=Rp}$$

Eq. (5) was solved numerically using a fourth-order Runge-Kutta algorithm with Brent's method to satisfy the boundary conditions (Press et al., 1989), in a similar manner as proposed by de Gooijer et al. (1991).

The change of the location-dependent increase in biomass concentration in the particle ($C_{X,P}$) is obtained through discretization:

$$C_{X,P}(r,t+\Delta t) = C_{X,P}(r,t) + R_{X,P}(r,t)\cdot\Delta t \tag{6}$$

With the volumetric rate of biomass production in the particle ($R_{X,P}$) defined as:

$$R_{X,P}(r) = \left\{\mu^{max}\cdot\frac{C_{S,P}(r)}{C_{S,P}(r)+K_S} - Ms\cdot Y_{XS}\left(1-\frac{C_{S,P}(r)}{C_{S,P}(r)+K_S}\right)\right\}\cdot C_{X,P}(r) \tag{7}$$

The biomass concentration in the reactor ($C_{X,R}$) is calculated by correcting for the number of particles (*PNo*) in the reactor as follows.

$$C_{X,R}(t+\Delta t) = \frac{\sum_{i=0}^{Rp} C_{X,P}(r,t+\Delta t)\cdot V_{PSeg}(r)}{V_R}\cdot PNo \tag{8}$$

Where, $V_{PSeg}(r)$ stands for the location in the biofilm-dependent volume of the particle segment, and $V_R$ refers to the total reactor volume. The (constant) number of particles (*PNo*) in the reactor is calculated according to:

$$PNo = \left.\frac{C_{XT,R}\cdot V_R}{C_{XT,P}\cdot V_{PT}}\right|_{t=0} \tag{9}$$

Where, $C_{XT,P}$ and $C_{XT,R}$ are the total biomass concentration in the particle and reactor, respectively, and $V_{PT}$ is the total volume of the particle.

Integration in time of the substrate and product concentrations in the system is obtained by introduction of the efficiency ($\eta_S$) for substrate conversion according to:

$$\eta_S = \frac{3\cdot\alpha\cdot D_S\cdot\left.\frac{dC_{S,P}(r)}{dr}\right|_{r=Rp}}{Rp\cdot R_{S,R}^{max}} \tag{10}$$

Where $R_{S,R}^{max}$ stands for the maximum substrate-conversion rate in the reactor in the absence of diffusion limitations. Introduction of the $\eta_S$ in the differential equations for description of substrate conversion and product formation in time, results in the following equations:

$$\frac{dC_{S,R}(t)}{dt} = \frac{Q}{V_R}\cdot(C_{S,B}(t)-C_{S,R}(t)) + R_{S,R}\cdot\eta_S \tag{11}$$

$$\frac{dC_{S,B}(t)}{dt} = \frac{Q}{V_B}\cdot(C_{S,R}(t)-C_{S,B}(t)) \tag{12}$$

$$\frac{dC_{M,R}(t)}{dt} = \frac{dC_{S,R}(t)}{dt} - R_{XT,R} \tag{13}$$

Where $Q$ is the flow rate, $C_{S,B}$ is the substrate concentration in the buffer or recirculation tank, $V_B$ is the volume of the



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buffer tank, and $R_{XT,R}$ was calculated using equations similar to Eqs. (7) and (8), without correction for biomass decay.

### *Computation Procedure*

The computation starts by introducing the parameter values, variables, and initial conditions described in Table I and proceeds as follows: (1) The substrate profile in the granule is calculated [Eq. (5)]. (2) Based on this substrate profile, the biomass concentration in each segment of the particle [Eq. (6)], as well as the biomass concentration in the reactor after one time step [Eq. (8)] is calculated. (3) The efficiency factor for substrate conversion is calculated [Eq. (10)]. (4) Then, the substrate and product concentrations in the reactor are integrated [Eqs. (11) to (13)] by using the biomass calculated in step 2. This results in a new substrate concentration in the reactor. (5) The computation returns to step 1, and the loop stops until the substrate in the reactor is depleted.

## RESULTS

### Transport Limitations in Anaerobic Granular Sludge

The kinetics of acetate degradation by dispersed and granular methanogenic sludge were studied. The initial substrate [$C_{S,R}$ (0)], and biomass concentration [$C_{X,R}$ (0)], as well as the apparent $K_S$-value were estimated from the measured methane-production rate as a function of time with Eqs. (1) to (3) (simple Monod model). The $C_{S,R}$ (0)-value needed to be estimated because not all measured methane data were used in the estimation procedure as indicated in Figure 2.

The apparent $K_S$-value represents the affinity for the substrate and strong changes on its value serve as an indication for substrate-transport limitations. The results from Figure 2 clearly show that the anaerobic intact granules were mass-transport limited. The measured apparent $K_S$-values for dispersed and granular sludge were 32 ± 10 and 208 ± 35 g COD/m$^3$, respectively. The particles of the dispersed sludge were smaller than 200 μm, while the diameter of the granules was (3.5 ± 0.3) × $10^{-3}$ m.

### Relative Importance of Internal and External Mass Transport

To determine the quantitative impact of respectively external and internal mass transfer, experiments were conducted at varying $V_{up}$ and granule-mean diameters. Experiments were conducted at variable $V_{up}$ or variable granules-mean diameter while keeping the other parameter constant. An example of the recorded methane-production rate curves of such experiments is shown in Figure 3. From the methane-production-rate curves obtained, the apparent $K_S$-value was calculated by using the $K_S$ as a fitting parameter. For this section, the $K_S$ calculations were performed with the simple Monod model (i.e., without reaction-diffusion kinetics).

**Table I.** Parameters, initial conditions, and variables in the reaction-diffusion model and the corresponding methods used for determination or literature source.

| | Units | Value | Source |
|---|---|---|---|
| Initial conditions | | | |
| $C_{S,R}$ (0) | g S-COD/m$^3$ | | Estimated with the Monod model using the measured methane production rate data |
| $C_{S,B}$ (0) | g S-COD/m$^3$ | | |
| $C_{X,R}$ (0) | g X-COD/m$^3$ | | |
| $C_{XT,P}$ (0) | g X-COD/m$^3$ | | Measured[a] |
| $C_{XT,R}$ (0) | g X-COD/m$^3$ | | Measured[b] |
| $C_M$ (0) | g $CH_4$-COD/m$^3$ | 0 | |
| Parameters | | | |
| $\mu^{max}$ | d$^{-1}$ | 0.2 | Measured |
| $K_S$ | g S-COD/m$^3$ | 30 | Measured |
| $M_S$ | g S-COD/g X-COD · d | 0.57 | Measured |
| $Y_{XS}$ | g X-COD/g S-COD | 0.03 | Handel and Lettinga (1994) |
| $D_S$ | m$^2$/d | 0.00013 | Flora et al. (1995) |
| $V_R$ | m$^3$ | 0.00016 | Measured |
| $V_B$ | m$^3$ | 0.005 | Measured |
| Variables | | | |
| $R_p$ | m | | Measured |
| $V_{PT}$ | m$^3$ | | Calculated based on *Rp* |
| $Q$ | m$^3$/d | | Measured |
| $k_l$ | m/d | | Calculated based on an empirical relation valid for Reynolds number between 0 and 3 × $10^5$ Hooijmans et al. (1990) |
| α | — | | |

[a] $C_{XT,P} = C^1_{XR} \cdot \rho_X \cdot \gamma_X$ where $C^1_{XR}$ is the biomass concentration per unit of weight of wet sludge (g X-VSS/g sieved sludge), $\rho_X$ is the density of sieved sludge (1040 × $10^3$ g sieved sludge/m$^3$ sieved sludge), and $\gamma_X$ is the degree of reduction of biomass (1.36 g X-COD/g X-VSS).

[b] $C_{XT,R} = MW_{XT} \cdot C^1_{XT} \cdot \gamma_X / V_R$ where $MW_{XT}$ is the amount of sieved sludge (g) added to the reactor.



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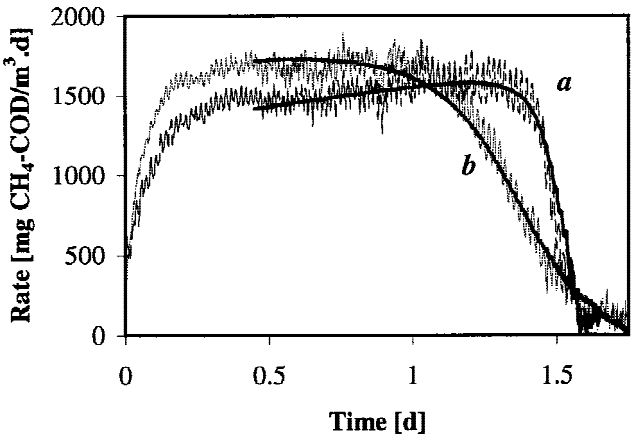



**Figure 2.** Measured (thin lines) and calculated (thick lines) methane-production rate of dispersed sludge (line *a*) and granular sludge (line *b*). The apparent $K_s$-values for the dispersed sludge and granules were 32 ± 10 and 208 ± 35 g COD/m$^3$, respectively. Note that for the estimation of the apparent $K_s$-values not all measured data were used.

### *The Effect of the Liquid Upflow Velocity*

The effect of different $V_{up}$ on the apparent $K_S$-value of granules with a mean diameter of (2.39 ± 0.09) × 10$^{-3}$ m was studied. The results show that increased $V_{up}$-values did not have an impact on the $K_S$-value of anaerobic granular sludge, suggesting that the effect of external mass transfer is limited (Fig. 4).

In subsequent feedings, a slight increase in the calculated $K_S$ was observed. The maximum substrate-consumption rate increases in time due to bacterial growth, which results in higher active biomass concentration in the granule. As a result of the increased substrate consumption rate, mass-transfer limitations become more important, and consequently, the value of the apparent $K_S$-value in time (e.g., subsequent feedings) increases.

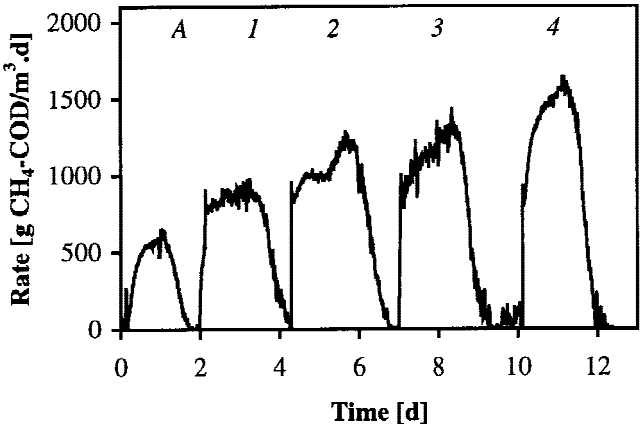



**Figure 3.** Example of on-line measurement of the methane-production rate. The run was performed with granules of (2.39 ± 0.09) × 10$^{-3}$ m and liquid-upflow velocity ($V_{up}$) of 3 m/h. During the tests, acetate was added at 2000 g S-COD/m$^3$ in four subsequent feedings (indicated by the numbers in the plot area). The letter "A" represents an acclimatization dose of ≈ 500 g S-COD/m$^3$.

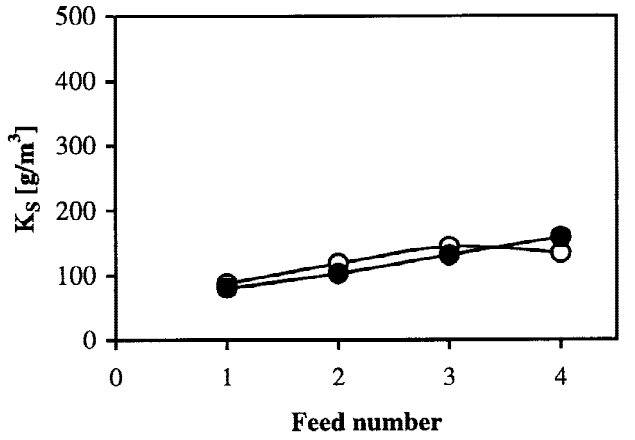



**Figure 4.** Apparent $K_s$-values of granular sludge with a mean diameter of (2.39 ± 0.09) × 10$^{-3}$ m, assessed at a $V_{up}$ of 3 m/h (○), and 10 m/h (●).

### *Effect of Different Granule Diameter*

The effect of the granule diameter on the apparent $K_S$ was studied at a $V_{up}$ of 10 m/h. The particle diameters tested were (2.39 ± 0.09) × 10$^{-3}$ m, (2.88 ± 0.26) × 10$^{-3}$ m, and (3.52 ± 0.28) × 10$^{-3}$ m. At increasing granule diameter, the substrate-consumption rate of the granules was decreased, as is reflected by an increase on the apparent $K_S$-value (Fig. 5). These results are clear indications for internal mass-transport limitations in anaerobic granular sludge.

## Biofilm Model Calculations

Using the data sets described above, we investigated if the biofilm theory could describe the experimental results (model validation). The procedure was the following: (1) The measured methane-production data were fitted by using the $K_S$-value as a fitting parameter (previous section). In this fitting procedure, the initial biomass and substrate concen-

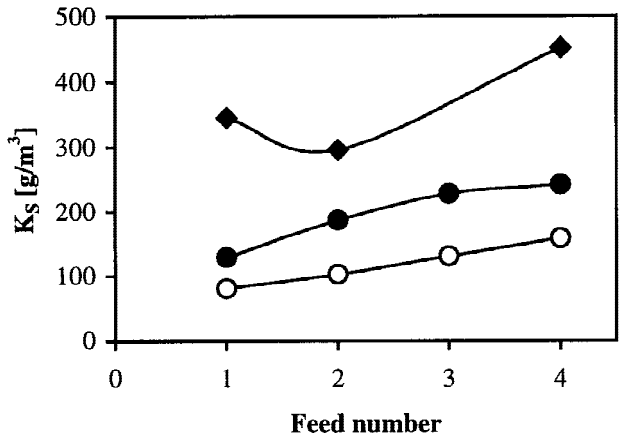



**Figure 5.** Apparent $K_s$-values at $V_{up}$ of 10 m/h for different biofilm sizes. The mean diameters of the granules were: (2.39 ± 0.09) × 10$^{-3}$ m (○); (2.88 ± 0.26) × 10$^{-3}$ m (●); and (3.52 ± 0.28) × 10$^{-3}$ m (◆). A technical problem resulted in incomplete data during the third feed of one of the experiments, and the $K_s$-value could not be calculated.



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trations were also estimated, and used as input for the biofilm model. (2) Next, the same data was fitted again, this time including the biofilm kinetic expressions in the model (i.e., using the biofilm model) and setting the $K_S$ constant at the value measured for dispersed sludge (30 g/m$^3$). Here, only the value of effective diffusion ($\alpha$) was used as a fitting parameter.

An example of the model-running sequences is shown in Figure 6. The biofilm model was able to relate the reaction-diffusion dynamics inside the granules with the measured parameters of the system and the measured $CH_4$ production-rate changes outside the biofilms. The $\alpha$ values for optimal description of the experimental data were similar for all the operational conditions tested (e.g., different granule diameter and different $V_{up}$). The values of $\alpha$ that resulted in an optimal description of the experimental results are shown in Table II.

### Sensitivity Analysis of the Biofilm Model

According to the film theory, the $V_{up}$ would have a positive impact on the external mass transfer by reducing the stagnant liquid layer around the granule. This means that, at higher $V_{up}$, the external mass-transfer coefficient ($k_1$) increases. On the other hand, the kinetic properties, and the size of the granule will be mainly related to the internal mass transport.

The relative importance of external and internal mass-transfer limitations can be theoretically estimated. For this purpose, a sensitivity analysis of the biofilm model was made around the set point values indicated in Table III. Each parameter was varied in the range 0.1–2.0 times the set-point value, keeping all other variables constant. The predicted efficiency for substrate conversion ($\eta_S$), or relative substrate-uptake rate at steady state was used for comparison.

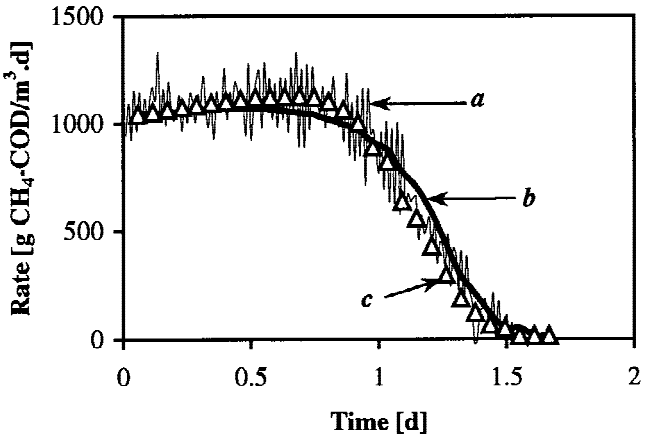



**Figure 6.** Analysis of the methane-production rate of granular sludge. The mean diameter of the granules was (2.88 ± 0.26) × 10$^{-3}$ m, and the upflow velocity was 10 m/h. Measured data (*a*). Fitting by varying the $K_s$-value (*b*). Fitting with the biofilm kinetic theory and varying the effective diffusivity $\alpha$ (*c*).

**Table II.** Values of the effective diffusivity $\alpha$ (−), that best described the methane-production rate curves.

| Diameter (m) | Upflow velocity (m/h) 3 | 10 | 15 |
|---|---|---|---|
| 2.39 × 10$^{-3}$ | 0.3 ± 0.1 | 0.3 ± 0.1 | |
| 2.88 × 10$^{-3}$ | | 0.4 ± 0.2 | 0.4 ± 0.2 |
| 3.52 × 10$^{-3}$ | | 0.3 ± 0.1 | |

**Table III.** Parameters used in the sensitivity analysis.

| Parameter | Set-point value | Units |
|---|---|---|
| $C_{S,R}$ | 50 | g S-COD/m$^3$ |
| $C_{X,P}$ | 10,000 | g X-COD/m$^3$ |
| $K_S$ | 30 | g/m$^3$ |
| $r$ | 0.001 | m |
| $k_l$ | 1 | m/d |
| $\alpha$ | 0.3 | — |

The simulations clearly show that $C_{S,R}$, $R_P$, and $C_{X,P}$ have a high impact on the relative substrate-uptake rate (Fig. 7a), while $K_S$, $k_1$, and $\alpha$ have a low impact (Fig. 7b). These simulation outputs will be discussed in comparison with the experimental results in the next two sections.

## DISCUSSION

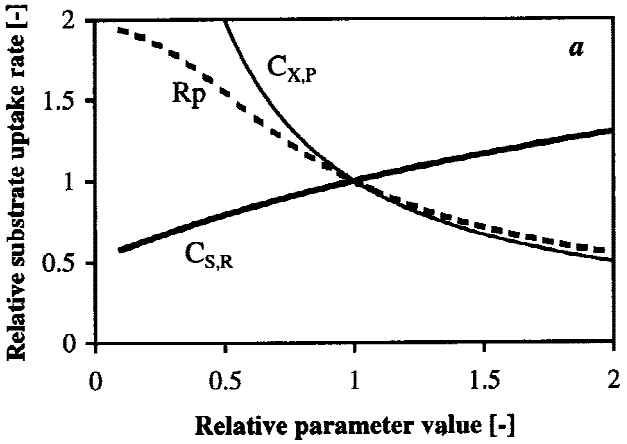
### External Mass Transport

The sensitivity analysis demonstrated that external mass-transfer limitations may play an important role in anaerobic granular biomass at $k_1$ values lower than 0.3 m/d (Fig. 7b). This $k_1$ value corresponds, according to an empirical relation valid for Reynolds from 0 to 3 × 10$^5$, to $V_{up}$-values lower than 1 m/h, suggesting that external mass-transfer limitations are not important under the conditions normally encountered in anaerobic bioreactors. There are several empirical relations to estimate $k_1$ values, however, independent of the empirical relation used, the $k_1$ values calculated for $V_{up}$ of 1 m/h were in all cases higher than 0.3 m/d (Fig. 8). We can conclude that the $V_{up}$ would hardly influence the substrate conversion rate of anaerobic granules at values higher than 1 m/h. This corresponds well with our experimental results that showed that at increasing values of $V_{up}$ the apparent $K_S$-values did not change. This confirms that external mass-transport resistance is normally not rate-limiting for anaerobic granular sludge.

According to the literature, external mass-transfer resistance can be neglected in denitrifying biofilters (Christiansen et al., 1995), in acetate-fed methanogenic fluidized bed (Wang et al., 1986), and theoretically the same is true in packed-bed reactors for aggregates up to 2 mm (Pavlostathis and Giraldo-Gomez, 1991). In other reports, it has been suggested that liquid-film mass-transfer resistance plays a strong role in substrate utilization in UASB reactor systems



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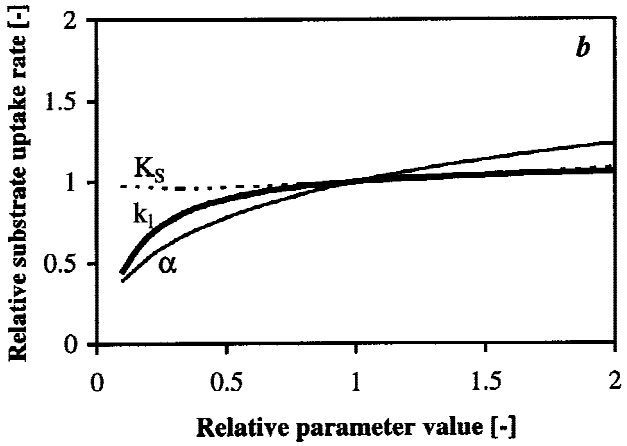




**Figure 7.** Parametric sensitivity analysis of the model. (*a*) High sensitive parameters. $C_{S,R}$ is the substrate concentration in the reactor, *Rp* is the radius of the particle, and $C_{X,P}$ is the biomass concentration in the particle. (*b*) Low-sensitive parameters. $K_S$ is the Monod half-saturation constant, $k_l$ is the external mass-transfer coefficient, and α is the relative effective diffusivity.

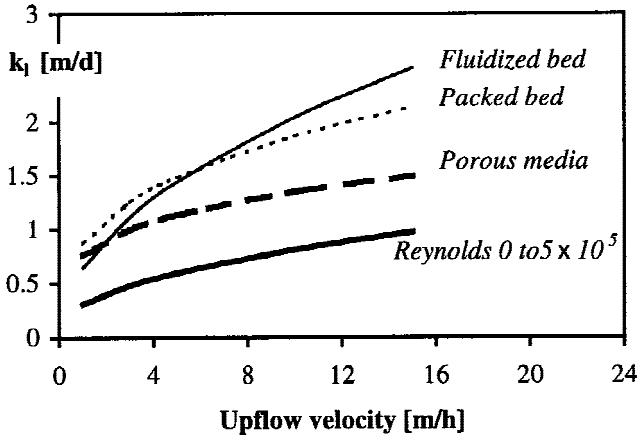



**Figure 8.** Calculated external mass-transfer coefficient-values ($k_l$), as a function of the upflow velocity ($V_{up}$). For the calculations several empirical relations were used. In all cases the diameter of the biofilm was $3.5 \times 10^{-3}$ m. The empirical relations to calculate the curves for fluidized and packed-bed systems were taken from Kissel (1986). The curve for porous media was computed as described by Rittman et al. (1986), while the empirical relations to calculate the curve valid for Reynolds from 0 to $3 \times 10^5$ were taken from Hooijmans et al. (1990).

(Brito and Melo, 1997; Wu et al., 1995). However, the reduction of liquid-phase mass-transfer resistance (Zaiat et al., 1996) and reduction on the $K_S$ values at increasing $V_{up}$-values (Kato, 1994) in these studies may be attributed to enhanced mixing of the liquid and the sludge bed. The effect of the $V_{up}$ may then be attributed to the reduction of preferential channeling of the influent wastewater and not to a direct effect on transport phenomena in the anaerobic biofilms. Further, it is important to consider that the gas production will contribute more than the $V_{up}$, to the mixing in reactors. The gas production per unit surface area is directly related to the height of the reactor so that in tall and thin reactors the gas load ($m^3/m^2$) can be very high. This is mainly true in full-scale compared to lab-scale systems.

## Internal Mass Transport

The simulations of the sensitivity analysis show that the substrate concentration in the reactor (i.e., bulk), the radius of the granule, and the biomass concentration have a high impact on the relative substrate-uptake rate (Fig. 7a). The experimental results also showed that, in contrast to the $V_{up}$, the diameter of the granules had a high impact on the apparent $K_S$. With increasing particle size, the $K_S$-value increased. This clearly indicates internal mass-transport limitations of the anaerobic sludge. Diffusion limitations have been reported for granular sludge from UASB systems (Dolfing, 1985; Wu et al., 1995). On the contrary, Alphenaar (1994) found no relationship between the measured $K_S$ and the granule diameter. However, in this study, a limited amount of biomass was active in the outer layer and the cores of the granules are reported to be empty. Schmidt and Arhing (1991) did not observe differences on acetoclastic methanogenesis of disintegrated and intact granules. Likely, the small size of the granules in their study ($0.44 \times 10^{-3}$ m) played a role in the results.

Although it was clearly shown that anaerobic biofilms could be mass transport limited, if we considered: (1) the substrate concentrations commonly found in wastewaters, (2) that the biofilm thickness experienced in most anaerobic systems is of mean diameters smaller than $1 \times 10^{-3}$ m (Pereboom and Vereijten 1994), and (3) the not very high-specific methanogenic activity on acetate ranging from 0.4 to 0.6 g $CH_4$-COD/g VSS · d, (Schmidt and Ahring, 1996) of most granular sludge treating different waste streams—this suggests that in practice, diffusion limitations might not be of extreme importance. Hence, the conversion of the rate-limiting substrate (i.e., acetate) will be mainly controlled by its biological kinetics.

Another aspect that should be considered in the biofilm conversion process is the formation of gaseous products. By assuming that (1) only molecular diffusion applies, (2) that the diffusion coefficient for acetate is equal to that of meth-



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ane ($CH_4$), and (3) that complete conversion of substrate into $CH_4$ occurs, then, it is possible to get an approximate estimation of the $CH_4$ profiles inside the granules. This estimation is shown in Figure 9. The substrate profile that consecutively led to the $CH_4$ profile was obtained as computed from one of the experimental data sets already described. It can be seen from Figure 9 that at high-granule diameters with a high active biomass concentration, steep substrate profiles, consequently, steep $CH_4$ profiles may develop. The high $CH_4$ concentration difference between the inner biofilm and the bulk liquid will create a high out-flux of $CH_4$. A gas with low-aqueous solubility like $CH_4$ may reach supersaturated concentrations inside the anaerobic granules and under supersaturated conditions $CH_4$ gas bubbles may be formed. It has been questioned how the formation and out-flux of the formed gas bubbles affects the activity of biofilms. In addition, it has become speculative if microcurrent movements can be generated due to bubble movements and whether those currents, in turn, may cause substrate transport by convection (Henze and Harremoes, 1983). Such a situation would result in insignificant diffusional resistance.

To assess if transport other than diffusion (e.g., convection) would play a role in the kinetics of biofilms, we used the parameter $\alpha$ as a tool. The value of $\alpha$ is expected to be rather constant for the same type of sludge, as long as it depends on the structure of the biofilm. The parameter $\alpha$ is defined as the ratio between the diffusion coefficient of substrate in the biofilm and in pure water. If the values of $\alpha$ that best describe the experimental results were different for different operational conditions (i.e., $V_{up}$ or granule diameter), it would indicate that diffusion in the biofilm is being affected by those operational conditions. Further, $\alpha$ values higher than 1 may also indicate that other types of mass transport (e.g., convection) may play a role in the conversion process. In this study there was no evidence of convective flow inside the anaerobic biofilms. Internal diffusion resistance was evident and was not eliminated by the gas produced during the tests or by increased $V_{up}$. The $\alpha$ value that explained the experimental results was similar for all the experimental conditions tested, and no indications of enhanced substrate transport could be found. The substrate uptake could be satisfactorily explained only by diffusion in the biofilm. This experimental result confirms the generally accepted assumption that diffusion is the predominant transport process that takes place in anaerobic biofilms (Pavlostathis and Giraldo-Gomez, 1991).

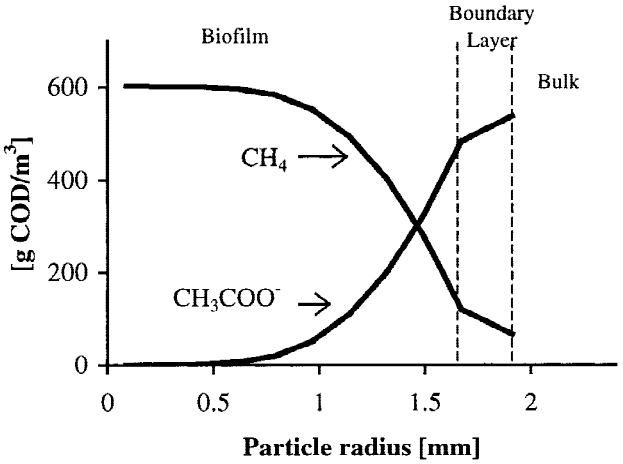



**Figure 9.** Acetate ($CH_3COO^-$), and methane ($CH_4$) profiles in a granule. The methane concentration at saturation was calculated considering 70% of $CH_4$ in the headspace. The acetate profile that led to the estimation of the methane profile was obtained from the experimental data set at a $V_{up}$ of 10 m/h, with granules of $3.5 \times 10^{-3}$ m, during a second feed run, and from a specific kinetic time.

We thank R. Wijffels for useful discussions.

## NOMENCLATURE

| | |
|---|---|
| $A^{max}$ | maximum growth rate ($d^{-1}$) |
| $C_{M,R}$ | methane concentration in the reactor (g $CH_4$-COD/$m^3$) |
| *COD* | chemical-oxygen demand |
| $C_{S,B}$ | substrate concentration in the buffer tank (g S-COD/$m^3$) |
| $C_{S,P}$ | substrate concentration in the particle (g S-COD/$m^3$) |
| $C_{S,R}$ | substrate concentration in the reactor (g S-COD/$m^3$) |
| $C^1_{XR}$ | biomass concentration per unit of weight of wet sludge (g X-VSS/g sieved sludge) |
| $C_{X,P}$ | active biomass concentration in the particle (g X-COD/$m^3$) |
| $C_{X,R}$ | active biomass concentration in the reactor, (g X-COD/$m^3$) |
| $C_{XT,P}$ | total biomass concentration in the particle (g X-COD/$m^3$) |
| $C_{XT,R}$ | total biomass concentration in the reactor (g X-COD/$m^3$) |
| $D_S$ | diffusion coefficient of substrate in water ($m^2$/d) |
| $f_{S,M}$ | COD-based stoichiometry for the conversion of substrate into product (methane) |
| $k_d$ | decay coefficient ($d^{-1}$) |
| $k_1$ | external mass transfer coefficient, (m/d) |
| $K_S$ | Monod half-saturation constant (g S-COD/$m^3$) |
| *Ms* | maintenance coefficient (g S-COD/g X-COD · d) |
| $MW_{XT}$ | amount of sieved sludge added to the reactor (g) |
| *PNo* | number of particles in the reactor |
| *Q* | recirculation flow rate ($m^3$/d) |
| *r* | radius (m) |
| *Rp* | radius of the particle (m) |
| $R_{M,R}$ | volumetric methane-production rate in the reactor (g $CH_4$-COD/$m^3$ · d) |
| $R_{S,P}$ | volumetric substrate-conversion rate in the particle, (g S-COD/$m^3$ · d) |
| $R_{S,R}$ | volumetric substrate-conversion rate in the reactor (g S-COD/$m^3$ · d) |
| $R^{max}_{S,R}$ | volumetric maximum substrate-conversion rate in the absence of diffusion limitations (g S-COD/$m^3$ · d) |
| $R_{X,P}$ | volumetric biomass growth rate in the particle (g X-COD/$m^3$ · d) |
| $R_{X,R}$ | volumetric biomass growth rate in the reactor (g X-COD/$m^3$ · d) |
| *t* | time |
| $V_B$ | volume of the buffer tank ($m^3$) |
| $V_{PSeg}$ | volume of the particle segment ($m^3$) |
| $V_{PT}$ | total volume of the particle ($m^3$) |
| $V_R$ | volume of the reactor ($m^3$) |
| *VSS* | volatile suspended solids |
| $V_{up}$ | liquid-upflow velocity (m/h) |
| $Y_{XS}$ | yield coefficient (g X-COD/g COD) |

*Greek letters*

| | |
|---|---|
| $\alpha$ | relative effective diffusivity (−) |



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| | |
|---|---|
| $\mu^{max}$ | maximum growth rate ($d^{-1}$) |
| $\eta_S$ | relative substrate-conversion efficiency (−) |
| $\gamma_X$ | degree of reduction of biomass (g X-COD/g X-VSS) |
| $\rho_X$ | density of sieved sludge (g sieved sludge/$m^3$ sieved sludge) |

## References


Alphenaar A. 1994. Anaerobic granular sludge: Characterization, and factors affecting its functioning. PhD Thesis, Wageningen Agricultural University, The Netherlands.

American Public Health Association (APHA). 1985. Standard methods for examination of water and wastewater, Washington DC: American Public Health Association.

Arcand Y, Chavarie C, Guiot SR. 1995. Dynamic modelling of the population distribution in the anaerobic granular biofilm. Wat Sci Technol 30:63–73.

Beeftink HH, van der Heijden RTJM Heijnen JJ. 1990. Maintenance requirements: Energy supply for simultaneous endogenous respiration and substrate consumption. FEMS Microbiol. Ecol 73:203–210.

Brito AG, Melo LF. 1997. A simplified analysis of reaction and mass transfer in UASB and EGSB reactors. Environ Technol 18:35–44.

Buffiere P, Steyer J-P, Fonade C, Moletta R. 1995. Comprehensive modeling of methanogenic biofilms in fluidized bed systems: Mass transfer limitations and multisubstrate aspects. Biotechnol Bioeng 48:725–736.

Christiansen P, Hollesen L, Harremoes P. 1995. Liquid film diffusion on reaction rate in submerged biofilters. Water Res 29:947–952.

de Beer de D, Huisman JW, van den Heuvel JC, Ottengraf SPP. 1992. The effect of pH profiles in methanogenic aggregates on the kinetics of acetate conversion. Water Res 26:1329–1336.

de Gooijer de CD, Wijffels RH, Tramper J. 1991. Growth and substrate consumption of *Nitrobacter agilis* cells immobilized in carrageenan: Part 1. Dynamic modeling. Biotechnol Bioeng 38:224–231.

Dolfing J. 1985. Kinetics of methane formation by granular sludge at low substrate concentrations. Appl Microbiol Biotechnol 22:77–81.

Flora JR, Suidan MT, Biswas P, Sayles GD. 1993. Modeling substrate transport into biofilms: Role of multiple ions and pH effects. J Environ Eng 119:908–930.

Flora JR, Suidan MT, Biswas P, Sayles GD. 1995. A modeling study of anaerobic biofilm systems: I. Detailed biofilm modeling. Biotechnol Bioeng 46:43–53.

Gupta N, Gupta SK, Ramachanran KB. 1997. Modelling and simulation of anaerobic stratified biofilm for methane production and prediction of multiple steady states. Chem Eng J 65:37–46.

Handel van AC, Lettinga G. 1994. Anaerobic sewage treatment: A practical guide for regions with a hot climate. Chichester: John Wiley & Sons Ltd.

Henze M, Harremoes P. 1983. Anaerobic treatment of wastewater in fixed film reactors. A litterature review. Wat Sci Tech 15:1–101.

Hooijmans CM, Geraats SGM, Potters JJM, Luyben AM. 1990. Experimental determination of the mass transfer boundary layer around a spherical biocatalyst particle. Chem Eng J 44:B41–B46.

Kato MT. 1994. The anaerobic treatment of low strength soluble wastewaters. Ph.D. Thesis, Wageningen Agricultural University, The Netherlands.

Kennedy KJ, Droste RL. 1986. Anaerobic fixed-film reactors treating carbohydrate watewater. Water Res 20:685–695.

Kissel JC. 1986. Modeling mass transfer in biological wastewater treatment processes. Wat Sci Technol 18:35–45.

Leenen EJTM, Boogert AA, Lammeren van AAM, Tramper T, Wijffels RH. 1996. Dynamics of artificially immobilized *Nitrosomonas europea:* Effect of biomass decay. Biotechnol Bioeng 55:630–640.

Pavlostathis SG, Giraldo-Gomez EG. 1991. Kinetics of anaerobic treatment. Wat Sci Tech 24:35–59.

Pereboom JHF, Vereijken TLFM. 1994. Methanogenic granule development in full scale internal recirculation reactors. Wat Sci Technol 30:9–21.

Press WH, Flannery BP, Teukolsky SA, Vetterling WT. 1989. Numerical recipes in Pascal: The art of scientific computing. Cambridge: Cambridge University Press.

Rittmann BE, Crawford L, Tuck CK, Namkung E. 1986. In situ determination of kinetic parameters for biofilms: Isolation and characterization of oligotrophic biofilms. Biotechnol Bioeng 28:1753–1760.

Schmidt JE, Ahring BK. 1991. Acetate and hydrogen metabolism in intact and disintegrated granules from an acetate-fed, 55°C, UASB reactor. Appl Microbiol Biotechnol 35:681–685.

Schmidt JE, Ahring BK. 1996. Granular sludge formation in upflow anaerobic sludge blanket (UASB) reactors. Biotechnol Bioeng 49: 229–246.

Tartakovsky B, Guiot SR. 1997. Modeling and analysis of layered stationary anaerobic granular biofilms. Biotechnol Bioeng 54:122–130.

Wang YT, Suidan MT, Rittman BE. 1986. Kinetics of methanogens in an expanded-bed reactor. J Environ Eng 112:155–170.

Wijffels RH, Gooijer de CD, Kortekaas S, Tramper J. 1991. Growth and substrate consumption of *Nitrobacter agilis* cells immobilized in carrageenan: Part 2. Model evaluation. Biotechnol Bioeng 38:232–240.

Wu MM, Criddle C, Hickey R. 1995. Mass transfer and temperature effects on substrate utilization by brewery granules. Biotechnol Bioeng 46: 465–475.

Zaiat M, Vieira LGT, Foresti E. 1996. Liquid-phase mass transfer in fixed-bed of polyurethane foam matrices containing immobilized anaerobic sludge. Biotechnol Techniques 10:121–126.




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